

microeconomics_2_L2

Subject: #microeconomics

Topic: Short-run, long-run competitive equilibrium, and social welfare.

Definition. The market demand is given by the function $D(p)$, which specifies the total quantity demanded at a given per-unit price p .

Note

We are given n firms with cost functions: $C_1(q), \dots, C_n(q)$. At the equilibrium point we assume, by definition, that the $C_i(q) = s_i(q)$.

Definition. On the supply side, with n firms, each with an individual supply curve $s_i(p)$, the **market supply** is the sum of all individual supplies:

$$S(p) = \sum_{i=1}^N s_i(p)$$

Tip

When drawing the aggregate price against quantity supplied / demanded, it is recommended to use the x axis for the variable.

Definition. The **market clearing price** p^* is the price that equalizes supply and demand. This concept is known as short run equilibrium:

$$S(p^*) = D(p^*)$$

Note

A price p^* satisfying this condition might not always exist.

In this short-run equilibrium, some firms may have $s_i(p^*) = 0$ and can have positive ($\pi > 0$) or negative ($\pi < 0$) profit.

Class example 2:

Market demand is given by: $D(p) = 20 - p$

Suppose that all $n > 1$ firms are identical (In the sense of all having the same cost function)

Suppose that a firm's cost function is:

$$C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$$

Note

We've seen this in **Class Example 1** so we know that:

$$S(q) = \begin{cases} 0 & p < 4 \\ 0 \text{ or } 2 & p = 4 \\ \frac{p}{2} & p > 4 \end{cases}$$

Find the short run (SR) equilibrium (Eq) (if it exists) for all n .

Case 1: Eq price is $p < 4$.

We can make the assumption that $D(p) > 16$. Furthermore, according to $S(q)$ we know that $S(q) = 0$. There is a demand of 16 units and a supply of 0 units. Thus, there is no equilibrium.

Case 2: Eq price is $p > 4$.

This means that $s_i(p) = \frac{p}{2}$. We are finding the SR Eq for all n firms, so: $S(q) = \sum_{i=1}^n = \frac{n \cdot p}{2}$.

We know that the condition the market clearing price p^* , which we need for the SR Eq. is $D(p^*) = S(p^*)$.

We solve; $20 - p^* = \frac{n \cdot p^*}{2} \Rightarrow p^* = \frac{40}{n+2}$. We know that $p^* > 4 \Rightarrow n < 8 \Rightarrow n \leq 7$

Case 3: Eq price is $p = 4$.

$S_i(4) = 0, S_i'(4) = 2, D(4) = 16$. Here we could have an equilibrium with 8 firms that produce 2 units each. Or:

$$\begin{cases} q_i^* = 2 & \text{first 8} \\ q_i^* = 0 & \text{next } n - 8 \end{cases} \quad \forall n \geq 8$$

Long-run Competitive Equilibrium

Assumption. All firms (of which we have an infinite supply of) are identical (they share the same cost structure), there are no sunk costs ($c(0) = 0$), and entry into or exit from the market is free.

Main Idea. The number of firms n is not fixed but adjusts in the long run through the process of entry and exit.

Definition. A **long-run equilibrium** consists of a market price p^* and a number of firms n such that:

1. Market supply equals market demand:

$$n \cdot S(p^*) = D(p^*)$$

2. Free entry and exit ensures that each firm makes zero economic profit:

$$\pi(s(p^*)) = 0$$

This "zero-profit" condition is equivalent to $p^* = AC(q^*)$, where q^* is the quantity supplied by each firm at price p^* .

Social Welfare: Demand and Inverse Demand

Note

Some might remember that demand functions can take more parameters than just price, such as income (or others i forgot), the reason we can currently disregard the other factors for our work is that those utility functions from micro 1 can be transformed to the standard maximization functions.

An example would be that $V(q, m) = V(q) + m \Rightarrow \max_{q \geq 0} V(q) - p \cdot q$.

Definition. For a consumer with quasilinear utility $v(q) + m$ (where $v'(q) > 0$, $v''(q) < 0$, and $v(0) = 0$), the solution to $\max_{q \geq 0} v(q) - p \cdot q$ yields the **inverse demand function**:

$$p(q) = mv(q) = D^{-1}(q)$$

This function represents the highest per-unit price a consumer is willing to pay for quantity q . An inverse demand function of $q = 10$ would be the highest price a customer would be willing to pay for 10 units of product.

Definition. The **demand function** $D(p)$ is the quantity a consumer wants to buy at a per-unit price p ($q = D(p)$).

Note

The total value, $v(q)$, a consumer gets from q units is the area under their inverse demand curve from 0 to q :

$$\int_0^q p(x)dx = \int_0^q mv(x)dx = v(q)$$

Social Welfare: Consumer, Producer, and Social Surplus

Definition. If q units are bought at a single per-unit price $p(q)$, the **Consumer Surplus (CS)** is the total value consumers receive minus the total amount they pay:

$$CS := \int_0^q p(x)dx - p(q) \cdot q$$

Definition. If q units are sold at a single per-unit price $p(q)$, the **Producer Surplus (PS)** is the total payment producers receive minus the total variable cost of production:

$$PS(q) := p(q) \cdot q - \int_0^q MC(x)dx$$

Note

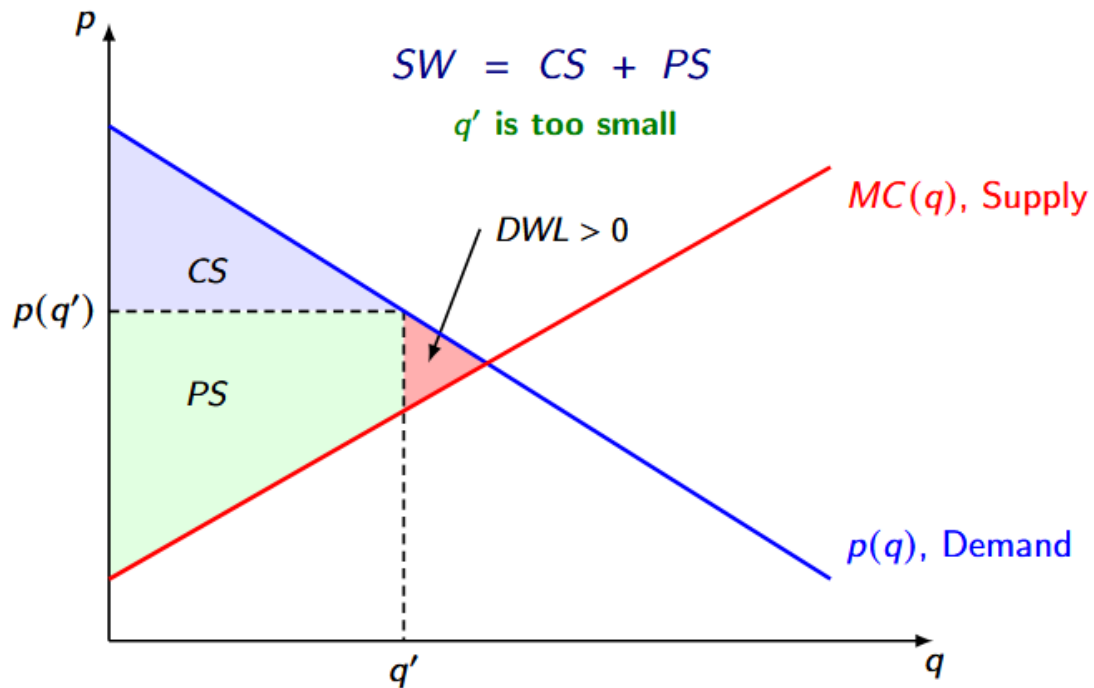
The sole difference between profit (π) and PS is that profit accounts for fixed costs, and producer surplus does not.

Definition. The sum of consumer and producer surplus is called **Social Welfare (SW)**:

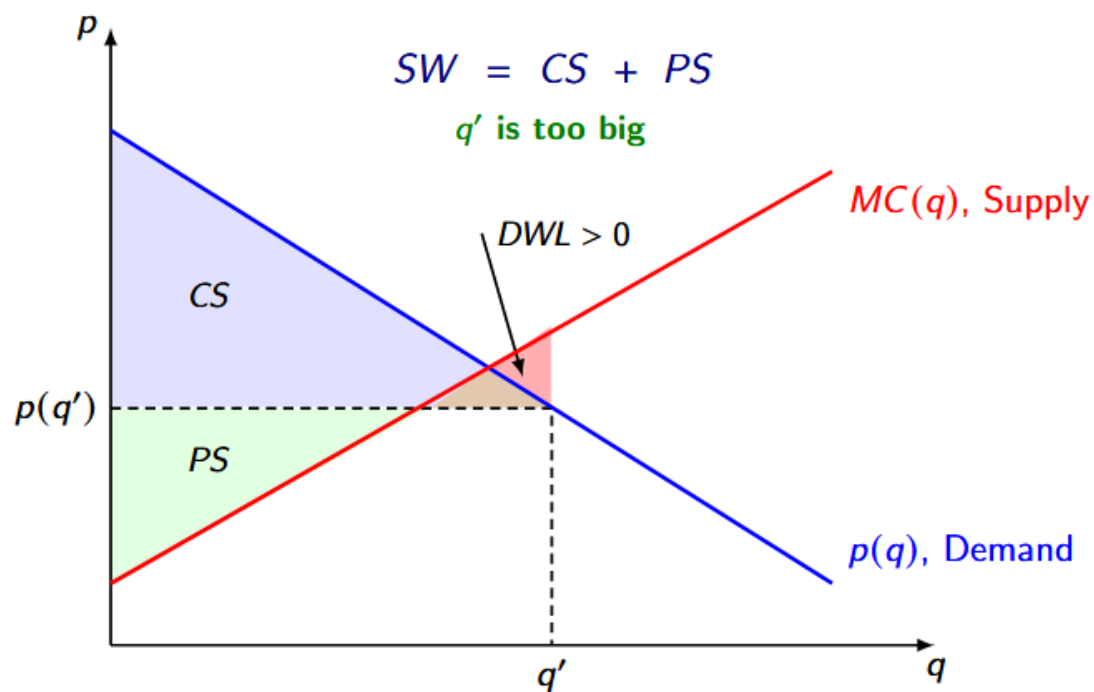
$$SW := CS + PS$$

This can be rewritten as:

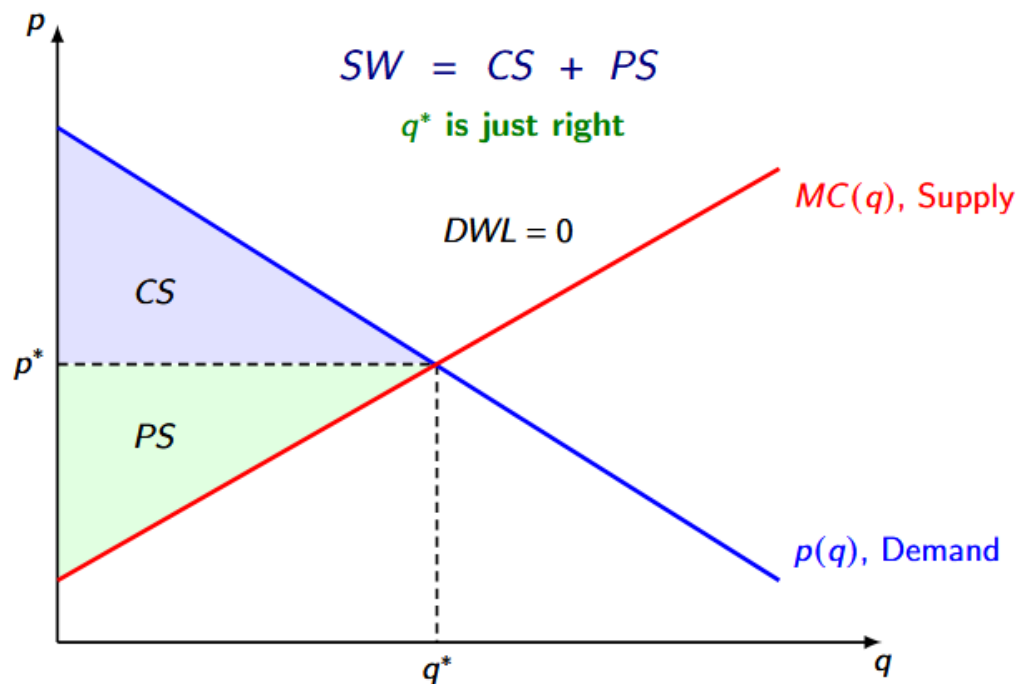
$$SW := \int_0^q p(x)dx - \int_0^q MC(x)dx$$



Too Little Output ($q' < q^*$): A deadweight loss ($DWL > 0$) exists because the value of additional units (given by the demand curve) exceeds the cost of producing them (given by the supply curve). Social welfare ($SW = CS + PS$) is not maximized.



Too Much Output ($q' > q^*$): A deadweight loss ($DWL > 0$) exists because the cost of producing the additional units exceeds their value to consumers. Social welfare ($SW = CS + PS$) is not maximized. The section where the producer takes on losses but the consumer wins out counts as a net zero for the social welfare.



This is why perfect competition is an ideal. It maximizes the social welfare.

Efficient Output (q^*): Social welfare ($SW = CS + PS$) is maximized. The market allocates resources efficiently, and the deadweight loss is $DWL = 0$.

Class Example 3

Market Demand: $D(p) = 30 - p$

Firm's Cost Function: $C(q) = q^2$ for $q \geq 0$

There are two ways to solve this task:

1. Finding the Competitive Equilibrium (p^*, q^*)

Set Supply equal to Demand: (equilibrium condition)

$$S(p^*) = D(p^*)$$

$$\frac{p^*}{2} = 30 - p^*$$

Solve for p^* :

$$\frac{p^*}{2} + p^* = 30$$

$$\frac{3p^*}{2} = 30$$

$$p^* = 20$$

Find q^* by plugging p^* into supply or demand:

$$q^* = S(p^*) = \frac{20}{2} = 10$$

$$q^* = D(p^*) = 30 - 20 = 10$$

Equilibrium: $(p^*, q^*) = (20, 10)$

2. Calculating Surpluses and Welfare

Marginal Cost and Inverse Functions

- Marginal Cost: $MC(q) = C'(q) = 2q$
- Inverse Demand Function: $p(q) = 30 - q$
- Inverse Supply Function: The supply curve is the MC curve above the shutdown point. Since $C(q) = q^2$ has $FC = 0$, the firm will supply at any $p > 0$. Thus, the inverse supply is $P = MC(q) = 2q$.

From this we can also obtain the equilibrium:

Remember that:

$$P(q) = MC(q)$$

Which is:

$$2q = 30 - q$$

Thus, equilibrium: $(p^*, q^*) = (20, 10)$

Now to find the CS and the PS, but this method is kind of cumbersome.

Tip

It is recommended to try graphing the supply and demand curves, which tends to make it easier to find the CS and PS without all of these calculations.

Consumer Surplus (CS)

CS is the area below the demand curve and above the price line.

$$\begin{aligned} CS &= \int_0^{q^*} (P(q) - p^*) dq = \int_0^{10} ((30 - q) - 20) dq \\ CS &= \int_0^{10} (10 - q) dq = \left[10q - \frac{1}{2}q^2 \right]_0^{10} \\ CS &= (100 - 50) - (0) = 50 \end{aligned}$$

Producer Surplus (PS)

PS is the area below the price line and above the supply (MC) curve.

$$\begin{aligned} PS &= \int_0^{q^*} (p^* - MC(q)) dq = \int_0^{10} (20 - 2q) dq \\ PS &= [20q - q^2]_0^{10} \\ PS &= (200 - 100) - (0) = 100 \end{aligned}$$

Social Welfare (SW) / Total Surplus

$$SW = CS + PS = 50 + 100 = 150$$

Deadweight Loss (DWL)

-In the perfect competition model, the equilibrium is Pareto efficient. Therefore, there is no deadweight loss.

$$DWL = 0$$

Sources:

Additional materials:

Review questions:

1. What does the demand function $D(p)$ represent in a competitive market?

2. How is the market supply $S(p)$ defined for n firms with individual supply functions $s_i(p)$?

3. What is the short-run competitive equilibrium?

4. Why can firms earn either positive or negative profit in the short-run equilibrium?

5. What two conditions define the long-run competitive equilibrium?

6. What is the key difference in assumptions between the short-run and the long-run?

7. What is the inverse demand function $p(q)$ and how is it interpreted?

8. How is consumer surplus (CS) calculated at equilibrium quantity q and price $p(q)$?

9. What is the difference between profit π and producer surplus (PS)?

10. What does deadweight loss (DWL) indicate in a competitive model?
